

A STRICTLY POSITIVE GLOBAL MINIMUM FOR THE RENORMALIZED QUARTIC OSCILLATOR

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ABSTRACT. Baez and Zhou reduced the self-consistent Wick ordering of a one-dimensional anharmonic oscillator to the shape of a single function of a single real parameter: writing $h_b = \frac{1}{2}(-d^2/dx^2 + bx^2) + x^4$ with normalized ground state G_b , the renormalized coupling map is $m(b) = b + 12\langle G_b, x^2 G_b \rangle$, and solutions at coupling $\lambda > 0$ correspond one to one with the real b solving $m(b) = \lambda^{-2/3}$. They proved $m > 0$ on the two parameter tails, computed m numerically in between, and conjectured that m has a strictly positive minimum. We prove the conjecture. A double-commutator identity evaluated in the ground state identifies $m(b)$ with a positive quadratic form on the orthogonal complement of G_b , which gives $m > 0$ at every real b without any restriction on the parameter; an explicit variational estimate makes m diverge at both ends of the parameter line, which confines the infimum to a compact interval and makes it attained. Consequently the critical coupling $\lambda_c = m_0^{-3/2}$ is finite and strictly positive, where $m_0 = \min m$.

1. INTRODUCTION

Wick ordering an interaction relative to its own ground state, rather than relative to the free vacuum, turns a renormalization prescription into a self-consistency problem: the state one orders against must be the state that the ordered Hamiltonian produces. Baez and Zhou [2] analysed this problem for the one-dimensional anharmonic oscillator, which is the simplest case in which it is not trivial, and reduced all of it to the shape of one function of one real parameter.

Their reduction is the following. For $b \in \mathbb{R}$ let

$$h_b = \frac{1}{2} \left(-\frac{d^2}{dx^2} + bx^2 \right) + x^4 \quad (1.1)$$

act on $L^2(\mathbb{R})$. The potential tends to $+\infty$, so h_b is bounded below with compact resolvent; its lowest eigenvalue $E(b)$ is simple and its normalized ground state G_b may be taken real, positive and even. Put

$$a(b) = \int_{\mathbb{R}} x^2 |G_b(x)|^2 dx, \quad m(b) = b + 12 a(b). \quad (1.2)$$

Baez and Zhou proved that the solutions of their self-consistency problem at coupling $\lambda > 0$ stand in one-to-one correspondence with the real b satisfying $m(b) = \lambda^{-2/3}$. The shape of m therefore decides both how many bare parameters produce a given renormalized coupling and, more sharply, for which couplings a solution exists at all. If m has a strictly positive minimum m_0 , then $\lambda_c = m_0^{-3/2}$ is the largest admissible coupling.

What [2] established about that shape left the middle of the parameter line open. For $b \geq 0$ positivity is immediate from $m(b) \geq b$, and for $b < -10$ their bound on the second moment gives $m(b) \geq -b/2 - 72/5$, which is positive once $b \leq -30$ and grows without bound as $b \rightarrow -\infty$; both tails were therefore already under control. On $[-30, 0]$ there was no bound. What they did there was to replace h_b by its Dirichlet restriction to $[-10, 10]$, whose own renormalized coupling they argue lies below m , and to compute that quantity by central differences with step 0.01; on the strength of the resulting graph they conjectured that m has a strictly positive minimum. The

gap was thus not the behaviour at infinity but positivity on a compact interval, and it was a gap because the only evidence there was a finite computation.

This paper closes that gap, and replaces the tail estimates by explicit ones valid on the whole of each half-line.

Theorem 1.1. *The function m of (1.2) is continuous on $\mathbb{R}$, satisfies $m(b) > 0$ for every $b \in \mathbb{R}$, and tends to $+\infty$ as $|b| \rightarrow \infty$. Consequently there exists $b_* \in \mathbb{R}$ with*

$$0 < m(b_*) = \min_{b \in \mathbb{R}} m(b).$$

Corollary 1.2. *Write $m_0 = \min m > 0$. The self-consistency problem of [2] is solvable at coupling $\lambda > 0$ if and only if $\lambda \leq m_0^{-3/2}$. In particular the critical coupling $\lambda_c = m_0^{-3/2}$ is finite and strictly positive.*

The proof of positivity is short and uses no property of the parameter, which is what removes the restriction to the tails. For any symmetric operator A preserving the Schwartz class and any ground state G of H at energy E ,

$$\langle G, [A, [H, A]]G \rangle = 2\langle AG, (H - E)AG \rangle. \quad (1.3)$$

Taking A to be the momentum turns the left side into the ground-state expectation of the second derivative of the potential, which for (1.1) is exactly $b + 12x^2$; that is, exactly $m(b)$. The right side is a quadratic form in AG , and because G_b is even while pG_b is odd, the vector pG_b lies in the orthogonal complement of the ground state, where the simplicity of $E(b)$ leaves a spectral gap. Positivity follows at every b at once. Identities of the type (1.3) are classical, and have long been used to extract moment information about anharmonic oscillators; what is used here is only that the right-hand side has a sign.

The operator family (1.1) is a dilation of the Montgomery family of quartic oscillators [13], which arose in the analysis of magnetic Schrödinger operators with a vanishing field and has since been studied both as a model for vortex nucleation [14] and in its own right [8, 7]; it belongs to the wider study of magnetic wells and magnetic bottles [10, 6], whose surrounding spectral theory is surveyed in [5], and the family reappears in the sub-Riemannian analysis of the Engel group [3]. The anharmonic oscillator itself has been studied since [4, 17] and remains a live subject [18].

One nearby result deserves to be separated from ours, because the two look interchangeable and are not. Helffer and Léautaud [9] proved that the lowest eigenvalue $\lambda_1(\alpha)$ of the Montgomery family has exactly one critical point. That theorem concerns the zeros of λ_1' , and the present question is a level set of λ_1'' . The dilation of Proposition 2.1 gives $E(b) = \lambda_1(-b/4) - b^2/16$, and Proposition 2.2 gives $m(b) = b + 24E'(b)$; differentiating the first and substituting it into the second,

$$m(b) = -2b - 6\lambda_1'(-\frac{b}{4}), \quad m'(b) = -2 + \frac{3}{2}\lambda_1''(-\frac{b}{4}), \quad (1.4)$$

so that $m'(b) = 0$ reads $\lambda_1'' = 4/3$, a level set of the second derivative rather than a zero of the first. Their result therefore neither implies nor is implied by anything proved here, and it is equally silent on the sign of m . It is also worth saying that the present theorem is about the sign and the attainment of m , and says nothing about how many critical points m has; that separate conjecture of [2] is not addressed.

Finally, the problem is a nonlinear eigenvalue problem in the sense that the operator depends on its own ground state, a setting with its own literature [12]; the reduction of [2] is what makes the present question a statement about one scalar function instead.

Section 2 collects what is needed about h_b and establishes the continuity of m . Section 3 proves positivity, Section 4 proves coercivity, and Section 5 assembles Theorem 1.1 and Corollary 1.2. Section 6 records what is not proved.

2. THE OPERATOR FAMILY AND THE CONTINUITY OF m

Throughout, $p = -i d/dx$, and

$$\mathcal{Q} = \{\psi \in H^1(\mathbb{R}) : x^2\psi \in L^2(\mathbb{R})\}$$

denotes the form domain.

Proposition 2.1. *For every $b \in \mathbb{R}$ the quadratic form of (1.1) is closed and bounded below on $\mathcal{Q}$, which is independent of b , and the associated self-adjoint operator has compact resolvent. Its lowest eigenvalue $E(b)$ is simple, and the corresponding normalized eigenfunction G_b may be chosen real, positive and even; moreover $G_b \in \mathcal{S}(\mathbb{R})$.*

Proof. For every $\varepsilon > 0$ there is C_ε with $|b|x^2/2 \leq \varepsilon x^4 + C_\varepsilon$, so the form $bx^2/2$ is infinitesimally form-bounded with respect to $p^2/2 + x^4$; the KLMN theorem [15] then gives a closed form bounded below on the b -independent domain $\mathcal{Q}$, and a unique associated self-adjoint operator. The potential tends to $+\infty$, so the resolvent is compact and the spectrum is a sequence of eigenvalues of finite multiplicity accumulating only at $+\infty$ [16].

The substitution $s = \sqrt{2}x$ together with $\alpha = -b/4$ carries h_b unitarily onto $M_\alpha - b^2/16$, where M_α is the Montgomery operator [13]. Simplicity of the lowest eigenvalue, the parity of its eigenfunction, membership in the Schwartz class and exponential decay of the eigenfunction and of all its derivatives are established for that family in [9, Proposition 1.1]; by the dilation they hold for h_b . The decay is of Agmon type, as for any Schrödinger operator whose potential grows at infinity [1]. Positivity of the ground eigenfunction in one dimension is Sturm's theorem. Consequently every pairing and every integration by parts used below is taken between Schwartz functions. $\square$

Proposition 2.2. *E is smooth on $\mathbb{R}$, and*

$$a(b) = 2E'(b), \quad \text{so that} \quad m(b) = b + 24E'(b) \quad (2.1)$$

is continuous on $\mathbb{R}$.

Proof. Smoothness of $\alpha \mapsto \lambda_1(\alpha)$ for the Montgomery family, together with the Feynman–Hellmann formula for it, is [9, Proposition 2.1]; the exact dilation of Proposition 2.1 gives $E(b) = \lambda_1(-b/4) - b^2/16$, so E is smooth. Since $b \mapsto h_b$ is a holomorphic family of type (B) on $\mathcal{Q}$ with $\partial_b h_b = x^2/2$, and $E(b)$ is a simple isolated eigenvalue, first-order perturbation theory [11] gives $E'(b) = \langle G_b, (x^2/2)G_b \rangle = a(b)/2$. Substituting in (1.2) yields (2.1), and continuity of m follows from smoothness of E . $\square$

3. POSITIVITY

Lemma 3.1. *Let H be self-adjoint and bounded below, with a normalized ground state $G \in \mathcal{S}(\mathbb{R})$ at energy E , and let A be symmetric and leave $\mathcal{S}(\mathbb{R})$ invariant, with AG in the form domain of H . Then*

$$\langle G, [A, [H, A]]G \rangle = 2\langle AG, (H - E)AG \rangle. \quad (3.1)$$

Proof. Because $HG = EG$ and H is symmetric on the vectors involved,

$$\begin{aligned} \langle G, AHAG \rangle - \langle G, A^2HG \rangle &= \langle AG, HAG \rangle - E\langle G, A^2G \rangle = \langle AG, (H - E)AG \rangle, \\ \langle G, AHAG \rangle - \langle G, HA^2G \rangle &= \langle AG, HAG \rangle - E\langle A^2G, G \rangle = \langle AG, (H - E)AG \rangle, \end{aligned}$$

where the second line used $\langle G, HA^2G \rangle = \langle HG, A^2G \rangle$. Adding the two identities and regrouping,

$$\langle G, (AHA - A^2H)G \rangle + \langle G, (AHA - HA^2)G \rangle = \langle G, [A, [H, A]]G \rangle,$$

since $[A, [H, A]] = A(HA - AH) - (HA - AH)A = 2AHA - A^2H - HA^2$. The left-hand side equals $2\langle AG, (H - E)AG \rangle$, which is (3.1). $\square$

Theorem 3.2. *$m(b) > 0$ for every $b \in \mathbb{R}$.*

Proof. Write $V_b(x) = \frac{1}{2}bx^2 + x^4$, so that $h_b = \frac{1}{2}p^2 + V_b$. Taking $A = p$ in [Lemma 3.1](#) and using $[p, [V_b, p]] = V_b'' = b + 12x^2$, which is the only term surviving because p commutes with p^2 , we obtain

$$m(b) = \langle G_b, (b + 12x^2)G_b \rangle = 2\langle pG_b, (h_b - E(b))pG_b \rangle. \quad (3.2)$$

The hypotheses of [Lemma 3.1](#) hold by [Proposition 2.1](#): G_b is Schwartz, hence so is pG_b .

Now G_b is even, so pG_b is odd and therefore orthogonal to G_b ; and $pG_b \neq 0$, since $pG_b = 0$ would force G_b to be constant, which is incompatible with $G_b \in L^2(\mathbb{R})$ and $\|G_b\| = 1$. By [Proposition 2.1](#) the eigenvalue $E(b)$ is simple and isolated, so there is $\delta = \delta(b) > 0$ with $h_b - E(b) \geq \delta$ on $\{G_b\}^\perp$. Hence

$$m(b) \geq 2\delta \|pG_b\|^2 > 0. \quad \square$$

Remark 3.3. The argument uses no information about b beyond what [Proposition 2.1](#) supplies, and in particular no numerical input. This is what extends positivity from the two tails, where [\[2\]](#) established it, to the whole parameter line.

4. COERCIVITY

Proposition 4.1. $m(b) \geq b$ for $b \geq 0$, and

$$m(-t) \geq 2t - 12\sqrt{\frac{t}{2} + 1} \quad \text{for every } t > 0. \quad (4.1)$$

In particular $m(b) \rightarrow +\infty$ as $|b| \rightarrow \infty$.

Proof. For $b \geq 0$ the second moment $a(b)$ is positive, so $m(b) = b + 12a(b) \geq b$.

Let $b = -t$ with $t > 0$. Completing the square in the potential,

$$h_{-t} = \frac{1}{2}p^2 + \left(x^2 - \frac{t}{4}\right)^2 - \frac{t^2}{16}. \quad (4.2)$$

Let $\psi_t(x) = \pi^{-1/4} \exp(-(x - \sqrt{t}/2)^2/2)$, a normalized Gaussian centred at the bottom of one of the two wells of [\(4.2\)](#). A direct computation gives $\langle \psi_t, \frac{1}{2}p^2\psi_t \rangle = 1/4$ and $\langle \psi_t, (x^2 - t/4)^2\psi_t \rangle = t/2 + 3/4$, so Rayleigh–Ritz applied to [\(4.2\)](#) yields

$$E(-t) + \frac{t^2}{16} \leq \frac{t}{2} + 1. \quad (4.3)$$

Evaluating [\(4.2\)](#) in the ground state G_{-t} and discarding the nonnegative kinetic term bounds the second moment of $x^2 - t/4$ by $E(-t) + t^2/16$, so by [\(4.3\)](#)

$$\langle G_{-t}, (x^2 - \frac{t}{4})^2 G_{-t} \rangle \leq \frac{t}{2} + 1.$$

The Cauchy–Schwarz inequality now bounds the first moment of $x^2 - t/4$ by the square root of its second moment,

$$\left| a(-t) - \frac{t}{4} \right| = \left| \langle G_{-t}, (x^2 - \frac{t}{4}) G_{-t} \rangle \right| \leq \sqrt{\frac{t}{2} + 1},$$

whence

$$m(-t) = -t + 12a(-t) \geq -t + 12\left(\frac{t}{4} - \sqrt{\frac{t}{2} + 1}\right) = 2t - 12\sqrt{\frac{t}{2} + 1}. \quad \square$$

Remark 4.2. The right-hand side of [\(4.1\)](#) becomes positive at $t = 9 + 3\sqrt{13} \approx 19.8$, the positive root of $t^2 - 18t - 36 = 0$, and grows linearly thereafter. Divergence on the left tail is not itself new: [\[2\]](#) bound the second moment for $b < -10$ and obtain $m(-t) \geq t/2 - 72/5$, which also grows. What [\(4.1\)](#) adds is a bound holding for every $t > 0$, with four times the slope, so that the interval on which neither bound of [Proposition 4.1](#) is positive is explicit and short. Positivity there is supplied by [Theorem 3.2](#), which uses no numerical input. [Figure 1](#) shows the two bounds and that interval.

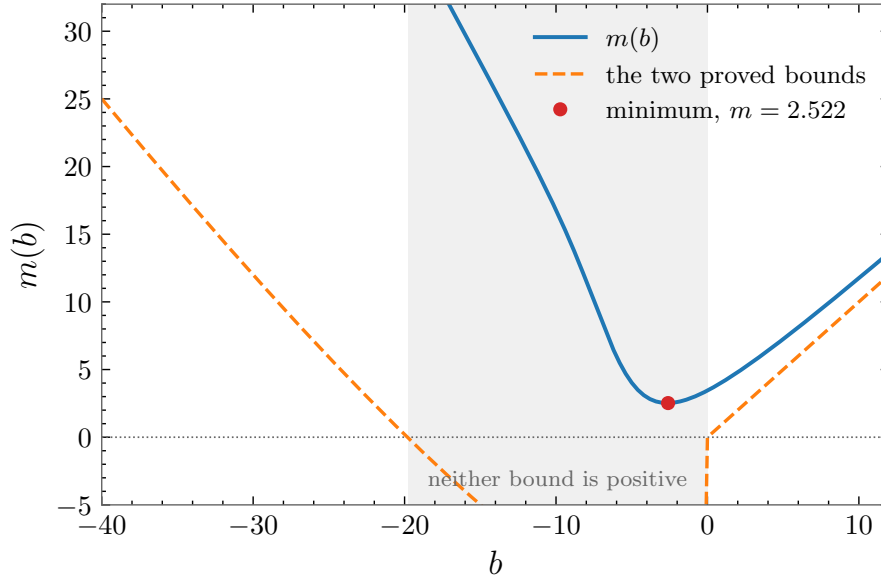


FIGURE 1. The two lower bounds of Proposition 4.1, $m(b) \geq b$ for $b \geq 0$ and (4.1) for $b < 0$, shown as dashed lines. They are positive outside the shaded interval, which confines any minimizer to a compact set; inside it, positivity comes from Theorem 3.2, which is independent of b . The solid curve is m itself, obtained by diagonalizing h_b in a harmonic-oscillator basis; it is drawn for orientation only and no statement in this paper depends on it.

5. THE MINIMUM AND THE CRITICAL COUPLING

Proof of Theorem 1.1. Continuity is Proposition 2.2, positivity is Theorem 3.2, and $m(b) \rightarrow +\infty$ as $|b| \rightarrow \infty$ is Proposition 4.1. For the last assertion, choose $R > 0$ so large that $m(b) > m(0)$ whenever $|b| > R$, which Proposition 4.1 permits. The restriction of m to the compact interval $[-R, R]$ is continuous and therefore attains a minimum at some $b_* \in [-R, R]$, and $m(b_*) \leq m(0) < m(b)$ for every $|b| > R$, so b_* minimizes m over $\mathbb{R}$. Positivity at b_* is Theorem 3.2. $\square$

Proof of Corollary 1.2. By [2], solutions at coupling $\lambda > 0$ correspond one to one with the real b satisfying $m(b) = \lambda^{-2/3}$. By Theorem 1.1 the range of m is contained in $[m_0, \infty)$ with $m_0 > 0$, and, m being continuous with $m(b) \rightarrow +\infty$ at both ends and attaining m_0 , its range is exactly $[m_0, \infty)$. Hence $m(b) = \lambda^{-2/3}$ has a solution if and only if $\lambda^{-2/3} \geq m_0$, that is, if and only if $\lambda \leq m_0^{-3/2}$. $\square$

6. WHAT IS NOT PROVED

Theorem 1.1 gives the existence of a minimizer and the strict positivity of the minimum value. It does not give uniqueness of the minimizer, and it does not address the separate conjecture of [2] that m has exactly one critical point; that question is a level set of λ_1'' under the dilation, and is not settled by [9], which settles the zeros of λ_1' .

Neither does the theorem give a numerical enclosure. The proof of Theorem 3.2 bounds $m(b)$ below by $2\delta(b)\|pG_b\|^2$, and both factors are accessible in principle; combined with the analytic dependence of the family on b and with interval arithmetic, this route would give certified rational-endpoint bounds for m_0 , and hence for $\lambda_c = m_0^{-3/2}$. That is not attempted here.

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